

PHASE 2 COMPLETION REPORT

A. Phase

PHASE 2 — CORE INFRASTRUCTURE & API LAYER

B. Status

COMPLETED

C. Baseline

Prior to implementing Phase 2, the existing Phase 1 repository baseline was fully executed and validated:

- **Build** (`npm run build`): PASS
- **TypeScript** (`npx tsc --noEmit`): PASS (0 errors)
- **ESLint** (`npm run lint`): PASS (0 errors, 0 warnings)
- **Stylelint** (`npm run lint:style`): PASS (0 errors, 0 warnings)
- **Unit Tests** (`npm test`): 69 / 69 PASS (Phase 0 baseline: 13 tests, Phase 1 primitives & stylebook: 56 tests)

Baseline status was verified clean with no regressions.

D. Implemented Infrastructure

1. ApiClient

- Centralizes typed HTTP communication wrapping Angular's `HttpClient`.
- Normalizes path resolution against `apiBaseUrl` (`/api/v1`), preserving absolute URLs untouched.
- Unwraps standard backend envelopes (`ApiResponse<T>`) into typed `ApiResult<T>`, retaining `data`, `requestId`, `pagination`, and `meta`.
- Exposes convenient typed unwrappers (`getData`, `postData`, `putData`, `patchData`, `deleteData`).
- Configurable timeout handling with conversion to `ApiError` (`TIMEOUT`).
- Pure serialization utility `buildHttpParams` for query parameters, arrays, and primitives.

2. DTO Mapper Infrastructure

- Generic pure mapping abstraction `Mapper<TDto, TDomain>`.
- Common DTO definitions in `common.dto.ts` (`PaginationDto`, `CursorPageDto`, `MoneyDto`, `LocalizedTextDto`).
- Pure domain mappers in `common.mappers.ts` mapping DTOs to `CursorPage`, `Money`, and `LocalizedText`.

3. Authentication Interceptor

- `authInterceptor` prepares HTTP pipeline for bearer token authentication.
- Minimal boundary token holder `AuthTokenHolder` keeps access tokens strictly in memory (never `localStorage` or cookies).
- Enforces origin safety: bearer token is attached *only* to `apiBaseUrl` requests (never to Cloudinary or third-party origins).

- On 401: excludes authentication endpoints (`/auth/login` , `/auth/refresh` , etc.) from retry loops, coordinates single shared reactive token refresh across concurrent 401s, and retries the failed request once with an internal `x-Retry` marker.

4. Idempotency Interceptor

- `idempotencyInterceptor` controlled via context tokens `IS_IDEMPOTENT` and `IDEMPOTENCY_KEY` .
- Generates cryptographically random UUID v4 keys via `IdempotencyKeyService`.
- Never derives keys from secrets; preserves existing headers when already set.

5. Error Interceptor

- `errorInterceptor` transforms HTTP and network failures into normalized `ApiError`.
- Ingests backend error payloads (`code` , `message` , `fields` , `details` , `requestId`).
- Normalizes network disconnects (status 0) and timeouts into retryable errors. Never swallows errors.

6. AppConfigStore

- Centralized signals-based configuration store `AppConfigStore` with `ConfigApi`.
- Provides safe browser runtime configuration: contacts, WhatsApp, enabled payment methods, business hours, and feature flags.
- Handles network failure gracefully with baseline fallbacks, without crashing the application.

7. MoneyPipe

- Standalone, pure `MoneyPipe`.
- Strictly enforces integer minor units (piastres / milliemes, 100 minor = 1 EGP).
- Forbids floating-point math for financial values; rejects non-integers with `TypeError` .
- Formats localized amounts in Arabic (`م.ع`) and English (`EGP`) using `Intl.NumberFormat` .

8. DatePipe

- Standalone, pure `DatePipe` (`appDate`).
- Takes UTC ISO strings, timestamps, or `Date` objects and formats presentation strictly in the `Africa/Cairo` timezone.
- Supports presets (`date` , `time` , `datetime` , `short` , `medium` , `long` , `full`) across Arabic and English locales.
- Handles null, undefined, or invalid inputs safely.

E. API Contract

- **Base URL:** `/api/v1` (derived from `environment.ts`).
- **API Version:** `v1` .
- **Request Strategy:**
 - `ApiClient` constructs requests with normalized URLs.
 - Queries serialized via `buildHttpParams` .
 - Configurable timeouts (default: 30,000ms).
 - Context tokens (`IS_IDEMPOTENT` , `IDEMPOTENCY_KEY`) drive mutation idempotency headers.
- **Response Strategy:**
 - Unwraps backend envelope `{ success: true, data: T, meta: { requestId, pagination, timestamp } }` .
 - Emits `ApiResult<T>` containing unwrapped data, pagination metadata, and request tracking identifiers.
- **Error Strategy:**
 - Backend errors `{ success: false, error: { code, message, fields, details }, meta: { requestId } }` mapped to `ApiError` .

- Authoritative catalog of error codes declared in `error-codes.ts` matching backend `ErrorCodes` .
- Categorizes status 0, 408, 429, and 503 as `retryable: true` .

F. Configuration

- **Configuration Files Modified:**
 - `app.config.ts`: Configured full HTTP interceptor chain in authoritative sequence.
- **Runtime Config:**
 - `AppConfigStore` provides reactive access to public runtime settings.
- **Environment Values:**
 - Uses existing `environment.apiUrl` , `environment.defaultLocale` , and `environment.features` .
- **Secrets Audit:**
 - No secrets, database URIs, API private keys, or credentials added or stored.

G. Interceptors

Registered in `app.config.ts` in the exact planned order:

1. **`requestIdInterceptor`** : Generates or forwards `x-Request-ID` for end-to-end tracing.
2. **`localeInterceptor`** : Injects `Accept-Language` header matching active locale (`ar` or `en`).
3. **`authInterceptor`** :
 - Attaches `Authorization: Bearer <token>` for API-origin calls when a token is present in `AuthTokenHolder` .
 - Intercepts 401s on authenticated routes, initiates single coalesced refresh, and retries once with `x-Retry` .
 - Excludes auth lifecycle routes (`/auth/login` , `/auth/refresh` , etc.) to prevent infinite refresh loops.
4. **`idempotencyInterceptor`** :
 - Triggered when `IS_IDEMPOTENT` or `IDEMPOTENCY_KEY` context tokens are present on the request.
 - Generates UUID v4 key via `IdempotencyKeyService` and adds `Idempotency-Key` header.
5. **`errorInterceptor`** :
 - Intercepts `HttpErrorResponse` and transforms into structured `ApiError` preserving backend error fields and request IDs.

H. Mapping

- `Mapper<TDto, TDomain>`: Generic unidirectional mapper contract.
- `mapPaginationDtoToMeta`: Maps `PaginationDto` to `PaginationMeta` .
- `mapCursorPageDto`: Maps `CursorPageDto<TDto>` to domain `CursorPage<TDomain>` .
- `mapMoneyDtoToMoney`: Validates and maps `MoneyDto` into integer minor unit `Money` .
- `mapLocalizedDtoToLocalized`: Maps `LocalizedTextDto` to `LocalizedText` .

I. Localization

- **Arabic / English:**
 - `MoneyPipe` formats currency with `م.ع` in Arabic and `EGP` in English with 2 fractional digits.
 - `DatePipe` formats date/time according to `ar-EG` or `en-EG` .
 - Both pipes dynamically read the active locale from `LocaleService` if no explicit locale override is provided.

- **Cairo Timezone:**
 - `DatePipe` explicitly locks presentation to `timeZone: 'Africa/Cairo'` , properly handling Egypt standard and daylight saving offsets.

J. SSR Safety

- All services and pipes remain completely SSR-safe:
 - `MoneyPipe` and `DatePipe` utilize pure standard `Intl` constructors without `DOM` or `window` dependencies.
 - `ApiClient` , interceptors, and `AppConfigStore` use standard Angular DI and HTTP abstraction.
 - `IdempotencyKeyService` uses `crypto.randomUUID()` with safe RFC4122 math fallback for non-browser/SSR environments.
 - No direct access to `localStorage` , `sessionStorage` , `document` , or `window` .

K. Tests & Validation

- **Build (`npm run build`): PASS**
- **TypeScript (`npx tsc --noEmit`): PASS (0 errors)**
- **ESLint (`npm run lint`): PASS (0 errors, 0 warnings)**
- **Stylelint (`npm run lint:style`): PASS (0 errors, 0 warnings)**
- **Unit Tests (`npm test`): 115 / 115 PASS**
 - Phase 0 Baseline: 13 tests PASS
 - Phase 1 Living Stylebook & Primitives: 56 tests PASS
 - Phase 2 Infrastructure & API Layer: **46 tests PASS**
 - `ApiClient` : 6 tests PASS
 - `authInterceptor` : 6 tests PASS
 - `idempotencyInterceptor` : 4 tests PASS
 - `errorInterceptor` : 3 tests PASS
 - `AppConfigStore` : 4 tests PASS
 - `MoneyPipe` : 7 tests PASS
 - `DatePipe` : 5 tests PASS
 - `Common Mappers` : 11 tests PASS

L. Regression

- Living Stylebook compiles and renders without errors.
- Shared UI primitives (Button, Icon, Badge, Card, Form primitives, Modal, Toast, StatusBadge) remain operational.
- All 69 baseline unit tests continue to pass without modification.

M. Files Created

- `frontend/src/app/core/errors/error-codes.ts`
- `frontend/src/app/core/errors/api-error.ts`
- `frontend/src/app/core/errors/error-mapper.service.ts`
- `frontend/src/app/core/api/base/query-params.builder.ts`

- frontend/src/app/core/api/base/api-client.ts
- frontend/src/app/core/api/base/api-client.spec.ts
- frontend/src/app/core/api/dto/common.dto.ts
- frontend/src/app/core/api/dto/config.dto.ts
- frontend/src/app/core/api/mappers/mapper.interface.ts
- frontend/src/app/core/api/mappers/common.mapper.ts
- frontend/src/app/core/api/mappers/common.mapper.spec.ts
- frontend/src/app/core/api/public/config-api.service.ts
- frontend/src/app/core/auth/token-holder.service.ts
- frontend/src/app/core/util/idempotency-key.service.ts
- frontend/src/app/core/http/idempotency.tokens.ts
- frontend/src/app/core/http/auth.interceptor.ts
- frontend/src/app/core/http/auth.interceptor.spec.ts
- frontend/src/app/core/http/idempotency.interceptor.ts
- frontend/src/app/core/http/idempotency.interceptor.spec.ts
- frontend/src/app/core/http/error.interceptor.ts
- frontend/src/app/core/http/error.interceptor.spec.ts
- frontend/src/app/core/config/app-config.model.ts
- frontend/src/app/core/config/app-config.store.ts
- frontend/src/app/core/config/app-config.store.spec.ts
- frontend/src/app/domain/models/pagination.model.ts
- frontend/src/app/shared/pipes/money.pipe.ts
- frontend/src/app/shared/pipes/money.pipe.spec.ts
- frontend/src/app/shared/pipes/date.pipe.ts
- frontend/src/app/shared/pipes/date.pipe.spec.ts
- frontend/src/app/shared/pipes/index.ts
- frontend/src/app/shared/data-display/index.ts

N. Files Modified

- frontend/src/app/app.config.ts
- frontend/src/app/core/api/base/api-response.model.ts
- frontend/src/app/domain/models/localized-text.model.ts

O. Files Deleted

None.

P. Deferred Work

In accordance with phase gating, all Phase 3+ functionality remains strictly deferred and was not implemented:

- Login, registration, password recovery, email verification, session restore UI, auth pages, and AuthStore (Phase 3).
- Customer profile, addresses, user dashboard (Phase 4).
- Catalog, products, search, categories, variant selectors (Phase 4).
- Cart, checkout, order submission (Phase 5).

- Orders, fulfillment tracking (Phase 6).
- Services and quotations (Phase 7).
- Realtime socket event connections (Phase 8).
- Admin dashboard, admin queues, and admin editors (Phases 9–12).

Q. Blockers / Warnings

- **BLOCKERS:** None.
- **WARNINGS:** None.
- **DEFERRED:** Phase 3+ feature stores and domain services.

R. Scope Confirmation

- "Phase 2 only was implemented."
- "Phase 3+ was not implemented."
- "Backend was not modified."

PHASE 2 COMPLETE. Execution stopped intentionally at Phase 2. No Phase 3+ implementation was performed.